

BANARAS HINDU UNIVERSITY

B.Sc. (Hons.) Semester III Sessional / Re-Mid Term Examination 2025-26 (NEP)

DEPARTMENT OF PHYSICS**Paper Code: PHYMN31 / PHYMJ31 / PHYMV31 : Applied Electronics
and Fiber Optics**

(Re-Mid Term Examination — Batch P2)

Time: 1 Hour

Maximum Marks: 20

Instructions: Answer all questions as specified. Figures to the right indicate full marks.

1. How does the refractive index difference between the core and cladding facilitate light transmission in optical fibers? [3]

OR

Why is single-mode fiber preferred for long-distance communication, while multimode fiber is used for shorter connections? [3]

2. If an optical fiber has a power loss of 5 dB, what percentage of the input power is lost? [3]
3. What are nonlinear losses in optical fibers and how do they differ from linear losses? [4]
4. A galvanometer has a resistance of $50\ \Omega$ and it deflects full scale when a current of 10 mA flows in it. How can it be converted into an ammeter of range 10 A? [3]
5. What do you mean by intrinsic and extrinsic semiconductor? [2]
6. An AC voltage source $V(t) = 200 \sin(100t)$ V is connected to a half-wave rectifier with a load resistance of $100\ \Omega$ (resistance of diode is $1\ \Omega$). Calculate: [5]
- (a) DC Power output,
 - (b) AC input power,
 - (c) Rectification efficiency
